

Nicholas Kantack

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Education

University of Oklahoma

Bachelor of Science: Engineering Physics
Graduated May 2016
Cumulative GPA: 3.98

University of Virginia

Master of Science: Electrical Engineering
Graduated December 2020
Cumulative GPA: 3.87

Work Experience

Amazon Web Services

July 2022 - Present

Software Development Engineer (L5)

- Backend development for Amazon Connect, a cloud-based customer service platform
- Principal architect for the rule-based and deep-learning-based algorithms powering Amazon Connect's new estimated wait time feature
- Developed anomaly detection algorithms, automation tools for DevOps, and led effort to migrate legacy infrastructure to AWS Fargate

Johns Hopkins Applied Physics Laboratory

July 2018 - July 2022

Software Developer/Electrical Engineer

- Principal investigator for six internal research grants totaling \$315,000 in funding
- Projects include AI in medicine, XR sensor fusion, state estimation for localization, and thin film RF device design

IntriCon Corporation

May 2016 - July 2018

Electronics Manufacturing Engineer

- Design and prototype sub-millimeter scale inductors for implantable medical devices
- Physical simulation and data analytics software design and support

Skills

Proficient	AWS (ECS, Lambda, Fargate, EC2, Bedrock, IAM, SageMaker, DynamoDB) Java, Javascript, Python, PyTorch, Docker, infrastructure as code, probability theory & continuous statistics, statistical tests & data analytics, modeling & simulation, calculus & differential equations, machine learning, neural networks, Android, Tampermonkey, OpenSCAD, Agile, 3D printing, digital circuits, VR/AR
Familiar	C++, C#, Tensorflow, CUDA, R, AutoCAD, Unity, Matlab, LaTeX, NodeJS, AngularJS, SQL, PCB design, RF systems, signal processing, NLP (HuggingFace transformers), Deep RL, robotics

Publications and Projects

Visit nickkantack.com for more information.